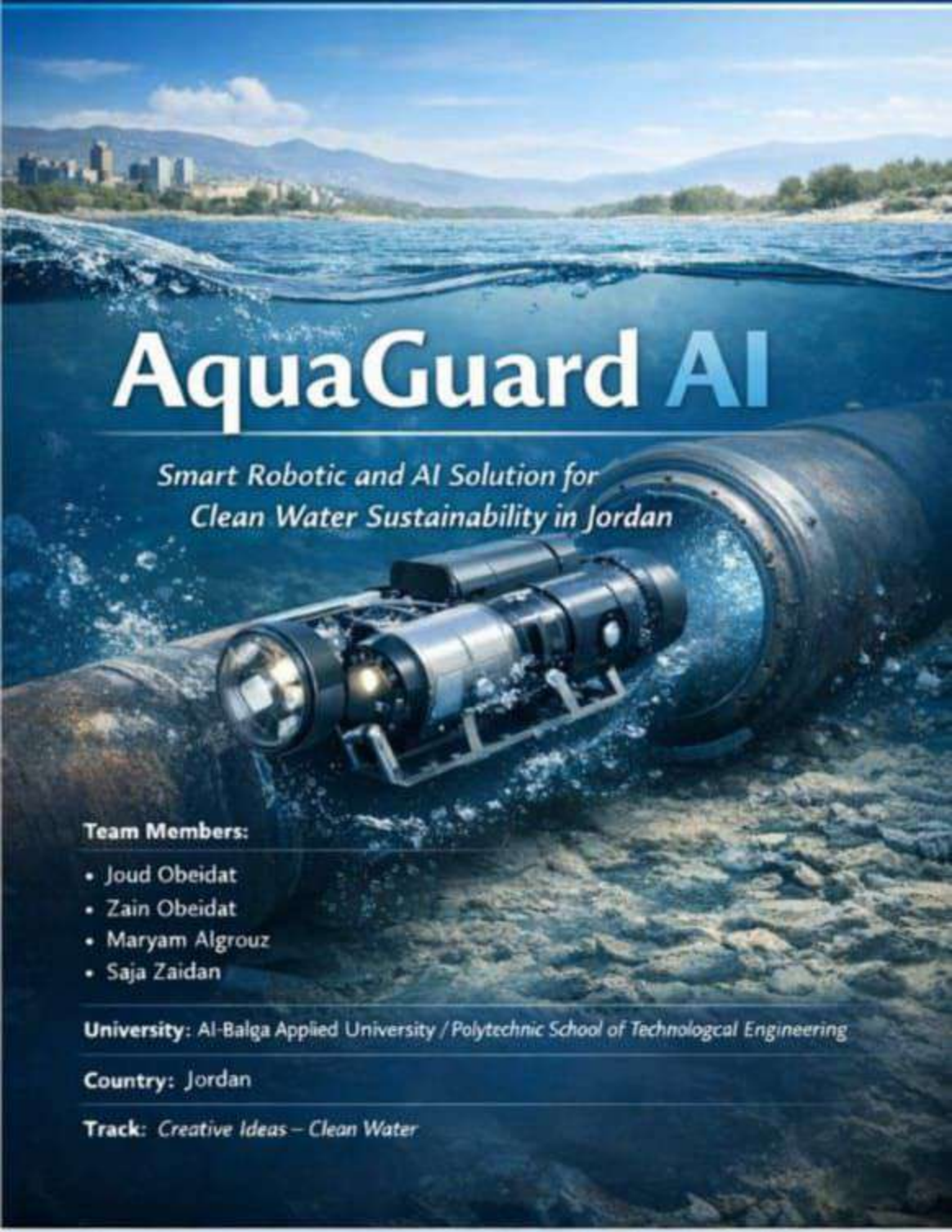


AquaGuard AI

Smart Robotic and AI Solution for Clean Water Sustainability in Jordan

Team Members:

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- Maryam Algrouz
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University: Al-Balqa Applied University / *Polytechnic School of Technological Engineering*

Country: Jordan

Track: *Creative Ideas – Clean Water*

Idea Title:
AquaGuard AI: Predictive Robotic Leak Detection for Water-Scarce Countries

Problem Description:
Jordan experiences significant water losses in its distribution networks due to undetected leaks and aging infrastructure. These non-revenue water losses reduce system efficiency, increase operational and maintenance costs, and put additional pressure on the country's limited water resources. Current inspection methods rely on manual checking or detection after major damage has occurred, leading to:
- Large volumes of wasted water
- Increased operational and maintenance costs
- Reduced availability of clean water for citizens
- Negative impacts on public health and sustainable urban development

Reducing water loss is essential to achieving the goal of clean water and sanitation in Jordan and ensuring sustainable water management.

Alignment with Sustainable Development Goals (SDGs):
- SDG 6: Clean Water and Sanitation (Primary Goal)
- SDG 3: Good Health and Well-Being
- SDG 11: Sustainable Cities and Communities
- SDG 13: Climate Action

Target Audience:
- National and local water authorities
- Municipalities and public/private water utilities
- Urban and semi-urban communities affected by water scarcit

Core Components:
1. Pipeline Inspection Robot
- Compact, pipe-compatible robot navigating inside pipelines
- Semi-autonomous navigation
- Equipped with acoustic and pressure sensors and internal cameras
- Collects real-time data while moving through the network

2. AI Analysis Engine
- Processes sensor data to detect early leaks and weak points
- Predicts potential failures before they occur
- Continuously improves accuracy using historical and real-time data

3. Smart Dashboard (User Interface)
- Displays leak locations, severity levels, and maintenance priorities
- Provides actionable insights for water utility managers

Innovation & Creativity:
- Integration of robotics with predictive AI instead of conventional late detection
- Shift from reactive maintenance to proactive, preventive strategies
- Gradual scalability without full infrastructure replacement
- Focus on intelligent water management rather than costly large-scale projects

Impact & Benefits:
- Environmental: Reduction in water loss, improved water efficiency
- Social: Increased access to clean water, improved public health
- Economic & Urban: Reduced maintenance costs, supporting sustainable cities
- Scalable & Transferable: Can be applied to other water-scarce regions

Robotics Component Details:
- Compact, pipe-compatible inspection robot
- Semi-autonomous navigation inside pipelines
- Equipped with acoustic sensors, pressure sensors, and internal cameras
- Water-safe materials and sealed electronics

Artificial Intelligence Component Details:
- Anomaly detection algorithms
- Predictive analytics
- Decision-support tools
- Continuous learning for improved accuracy

Required Data Sources:
- Real-time sensor data
- Historical pressure and flow records
- Maintenance and repair logs
- Environmental and seasonal water consumption data

Technical Feasibility & Safety:
Relies on proven robotic platforms and mature AI algorithms. Human supervision ensures safe and reliable operations, with ethical use of all collected data.

Impact, Measurability & Scalability:
- Impact: Reduced water loss and improved water security
- Measurable Indicators: % reduction in non-revenue water, early leak detection, maintenance cost savings
- Scalability: Adaptable to different city sizes and transferable to other regions

Cost Estimation:
Low to medium cost for prototype; modular design allows cost-effective scaling; cheaper than full pipeline replacement projects.

Idea Readiness Level:
TRL 3–4: Concept and architecture fully defined, ready for prototype and testing.

Risk & Ethical Considerations:
Secure data handling, safe robotic operations, transparent AI decision-making, ethical use of data.

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Potential Supporting Organizations:
- Government: Ministry of Water and Irrigation (MWI), Ministry of Environment
- Academic / Research: Jordan University of Science & Technology (JUST), University of Jordan
- NGO / International: UNDP Jordan, FAO Jordan, USAID Jordan
- Private / Technology Companies: Iris AI, Emkan, WaterTech companies

Final Statement:
Water loss is a silent crisis in Jordan. AquaGuard AI provides a creative, feasible, and high-impact solution to protect water resources using robotics, AI, and smart controls. By turning pipelines into intelligent systems, every drop of water is efficiently used and safeguarded for current and future generations.